

Cross-Market Coherence between Bitcoin and Nasdaq Futures

Magnitude-conditioned state persistence without incremental return predictability

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Abstract

This study asks whether contemporaneous Bitcoin and Nasdaq futures candle bodies define persistent cross-market states and whether relative move magnitude adds economically useful information. One-minute BTCUSDT and stitched NQ futures bars are synchronized in UTC, filtered with DST-aware exchange sessions, and aggregated to a five-minute primary representation. Directional coherence is measured over fixed 15-, 30-, and 60-minute clocks. Body magnitude is normalized by a causal same-slot median absolute deviation estimated from 63 prior eligible sessions. A purged, expanding walk-forward design compares a coherence-only logistic state model with a magnitude-conditioned extension and separately compares NQ-only and cross-market Ridge return forecasts. Magnitude reduces five-minute out-of-fold Brier loss by 6.99×10^{-4} , with a complete-session bootstrap 95% interval of $[-1.120, -0.297] \times 10^{-3}$. However, the cross-market return model increases mean squared error by 0.1924 bps^2 relative to NQ-only, with interval $[0.0728, 0.3222]$. The state result repeats overnight. The evidence therefore supports a narrow conditional-state association, not incremental NQ return predictability or tradable alpha.

Keywords: Bitcoin; Nasdaq futures; cross-market dependence; directional coherence; walk-forward validation; calibration; data leakage.

JEL classification: C53; G14; G17.

Research status. Completed development study, protocol v1.0.0. The record was locked after results for reproducibility; it is not a preregistration and no pristine final holdout was opened. All reported predictions are out of fold. Raw proprietary NQ data are not distributed.

1 Introduction

Bitcoin and Nasdaq futures often print candle bodies in the same direction while their volatility-adjusted magnitudes differ. That observation is contemporaneous: it does not by itself imply that BTC leads NQ, that NQ leads BTC, or that the weaker market must catch up. The empirical question is instead whether synchronized direction defines a persistent state, whether the scale and imbalance of the joint move help predict that state’s next transition, and whether any such information survives as an executable future NQ return.

The distinction matters. A model can improve the probability forecast for the next same-direction state without improving a conditional-mean return forecast. Likewise, a statistically stable state transition can be too small or too indirect to support a trading rule. This paper therefore separates three estimands: a descriptive response curve, a binary state-persistence forecast, and an incremental NQ return forecast.

The contribution is methodological rather than a claim of alpha. The analysis enforces next-open timing, causal normalization, fold-local preprocessing, nested regularization, complete-session purging, block-bootstrap inference, and a mandatory overnight negative control. Trial ledgers and compact aggregate outputs are retained so that unfavorable comparisons cannot be silently discarded. This follows the broader concern that repeated selection can make backtest performance materially overstate evidence [1, 2, 5].

2 Research questions and estimands

Let O_t^i and C_t^i denote the open and close of asset $i \in \{\text{BTC}, \text{NQ}\}$ during bar t . The signed candle-body return is

$$b_t^i = \log\left(\frac{C_t^i}{O_t^i}\right). \quad (1)$$

Wicks and high–low ranges are excluded from the primary definition. Per-bar directional agreement is

$$d_t = \text{sign}(b_t^{\text{BTC}}) \text{sign}(b_t^{\text{NQ}}) \in \{-1, 0, +1\}. \quad (2)$$

The study addresses two primary questions. First, conditional on $d_t = +1$, do magnitude features improve the forecast of $d_{t+1} = +1$ beyond recent directional coherence? Second, do BTC and coherence variables improve the forecast of the executable next five-minute signed NQ return beyond an otherwise comparable NQ-only information set? Longer horizons and the one-minute representation are prespecified secondary analyses.

3 Data and sample construction

The available sample combines Binance BTCUSDT perpetual one-minute OHLCV with a local stitched NQ front-month series. The common verified range is 7 March 2024 through 12 May 2026. Timestamps are stored in UTC and mapped to `America/New_York` for session logic. The cash-session sample follows the official 09:30–16:00 New York equity session, including early closes. The overnight control covers 18:00 on the prior evening through 09:30, excluding the CME maintenance interval, weekends, holidays, gaps, and contract-transition sessions.

The data begin after US spot Bitcoin ETP trading and were inspected in earlier project iterations. Consequently, the motivating post-ETP integration narrative is not tested as a causal before/after design, and this sample cannot provide a pristine final holdout. These restrictions are part of the inferential result rather than implementation details.

Table 1: Data provenance and eligibility constraints

Input	Available coverage	Principal constraint
BTCUSDT perpetual	2024-03-07 to 2026-05-12	Public Binance archive; post-ETP sample only
NQ stitched front month	Common range through 2026-05-12	Partial contract provenance; historical roll rule cannot be reconstructed
Cash sessions	458 complete sessions before warm-up	Entire incomplete or transition sessions excluded
Overnight sessions	551 complete sessions before warm-up	Separate negative control; cannot replace cash primary

Notes: Prices are never forward-filled. Five-minute bars require all constituent one-minute observations. A 63-session causal warm-up is applied after eligibility.

4 Feature construction

4.1 Directional coherence

For clock-time window $K \in \{15, 30, 60\}$ minutes, coherence is the unweighted average of complete agreement observations,

$$\mathcal{C}_t(K) = \frac{1}{n_K} \sum_{j=0}^{n_K-1} d_{t-j}, \quad \mathbf{C}_t = [\mathcal{C}_t(15), \mathcal{C}_t(30), \mathcal{C}_t(60)]. \quad (3)$$

The windows remain continuous on $[-1, 1]$ and are used jointly. No historically best window or coherence threshold is selected. The same clocks correspond to 3/6/12 bars in the five-minute representation and 15/30/60 bars in the one-minute robustness analysis.

4.2 Causal body scale and magnitude coordinates

Let $\mathcal{H}_t^{i,r}$ contain the previous 63 eligible body returns for asset i , resolution r , and the same DST-aware session slot as t . The current session is excluded. Define

$$\tilde{b}_{t-1}^{i,r} = \text{median}_{x \in \mathcal{H}_t^{i,r}} x, \quad (4)$$

$$\hat{\sigma}_{t-1}^{i,r} = 1.4826 \text{median}_{x \in \mathcal{H}_t^{i,r}} |x - \tilde{b}_{t-1}^{i,r}|, \quad (5)$$

$$m_t^i = \frac{|b_t^i|}{\hat{\sigma}_{t-1}^{i,r}}. \quad (6)$$

An observation is ineligible when fewer than 63 valid references exist or the scale is zero or non-finite. There is no epsilon floor or future-data fallback.

Joint intensity and relative magnitude balance are

$$J_t = \sqrt{m_t^{\text{BTC}} m_t^{\text{NQ}}}, \quad B_t = \frac{m_t^{\text{BTC}} - m_t^{\text{NQ}}}{m_t^{\text{BTC}} + m_t^{\text{NQ}}}. \quad (7)$$

J_t is symmetric and becomes large only when both markets move strongly relative to their own histories. $B_t \in [-1, 1]$ separates relative strength from common event scale; positive values indicate a relatively larger BTC body.

Table 2: Frozen parameter provenance

Parameter	Value	Role and status
Primary bars	5 minutes	Tractable primary representation; not claimed optimal
Robustness bars	1 minute	Mandatory timing check with the same economic endpoint
Coherence clocks	15/30/60 minutes	Multiscale state vector; no outcome-based best-window selection
Historical scale	63 sessions	Causal same-slot MAD; current session excluded
Economic horizon	5 minutes	Next-open to horizon-open at both resolutions
Purge	1 complete session	Exceeds the target horizon and separates adjacent folds

5 Prediction design and inference

5.1 Causal timing and outcomes

Features use information through the event-bar close τ_t . Economic measurement begins at the first tradable open after signal formation and ends at a future open:

$$R_{t,h}^{\text{NQ}} = \log\left(\frac{O_{\tau_t+h}^{\text{NQ}}}{O_{\tau_t}^{\text{NQ}}}\right), \quad Y_{t,h} = S_t R_{t,h}^{\text{NQ}}, \quad (8)$$

where S_t is the common detected direction and $h = 5$ minutes is primary. Thus positive $Y_{t,h}$ denotes continuation and negative values denote reversal. Horizons 15, 30, and 60 minutes form a secondary response curve; none can replace the five-minute endpoint.

5.2 Nested model comparisons

Conditional on $d_t = +1$, the state models are logistic regressions,

$$M_0 : \Pr(d_{t+1} = +1 \mid \mathbf{C}_t, S_t), \quad (9)$$

$$M_1 : \Pr(d_{t+1} = +1 \mid \mathbf{C}_t, S_t, J_t, B_t, |B_t|). \quad (10)$$

The primary state score is Brier loss; log loss and fixed-width reliability bins are secondary. The return comparison uses matched Ridge models. The NQ-only information set contains current NQ direction and magnitude, NQ momentum at 5/15/30/60 minutes, intraday Fourier terms, and weekday controls. The cross-market model adds BTC momentum and magnitude, the coherence vector, J_t , B_t , and $|B_t|$.

Both comparisons use expanding outer folds with 130 initial feature-ready sessions, one complete purged session, and 12 consecutive validation blocks of 22 sessions. All preprocessing and regularization selection occur within folds. Logistic $C \in \{0.01, 0.1, 1, 10\}$ and Ridge $\alpha \in \{0.01, 0.1, 1, 10, 100, 1000\}$ are selected from three purged inner expanding blocks. Every scored prediction is out of fold.

Paired uncertainty resamples complete validation sessions 5,000 times. For model loss L , the reported incremental comparison is

$$\Delta L = L_{\text{cross}} - L_{\text{baseline}}. \quad (11)$$

Negative values favor the expanded model. This convention is used for both the Brier and return-loss comparisons.

6 Empirical results

6.1 Descriptive response

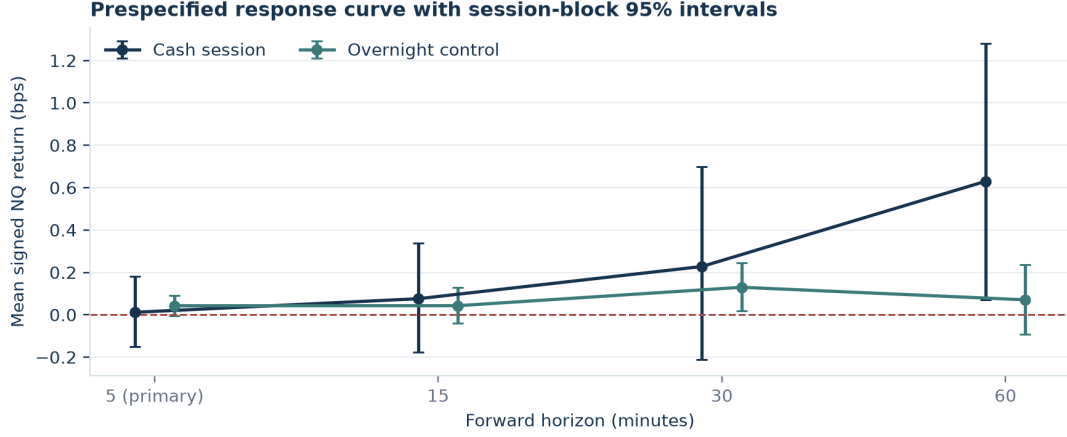


Figure 1: Signed NQ open-to-open response conditional on current BTC-NQ body-direction agreement. Intervals are 95% complete-session bootstrap intervals. The five-minute cash-session endpoint is primary; longer horizons and overnight estimates are secondary.

The cash-session five-minute descriptive estimate is 0.012 bps across 19,955 events, with an interval containing zero. The positive longer-horizon point estimates are much less precise and do not alter the registered primary conclusion. The unconditional response therefore supplies no trading premise before model comparison.

6.2 State persistence

The five-minute state sample contains 13,724 out-of-fold events from 264 validation sessions. The observed next-bar agreement rate is 68.25%. Table 3 shows that adding magnitude reduces both Brier and log loss.

Table 3: Five-minute out-of-fold state-prediction results

Model	Brier loss	Log loss	ECE
Fold training-rate benchmark	0.217250	0.626229	0.009849
M_0 : coherence	0.216736	0.625038	0.010755
M_1 : coherence + magnitude	0.216037	0.623370	0.009218
$M_1 - M_0$	-0.000699	-0.001668	-0.001537
95% block interval	[-0.001120, -0.000297] [-0.002668, -0.000703]		—

Notes: ECE denotes expected calibration error. Lower values are better. The Brier reduction is approximately 0.32% relative to M_0 .

M_1 improves Brier loss in 10 of 12 outer folds. Standardized coefficient signs are stable: mean coefficients are approximately +0.10 for joint intensity, -0.06 for signed balance, and +0.03 for absolute imbalance. These are conditional predictive associations, not structural or causal effects.

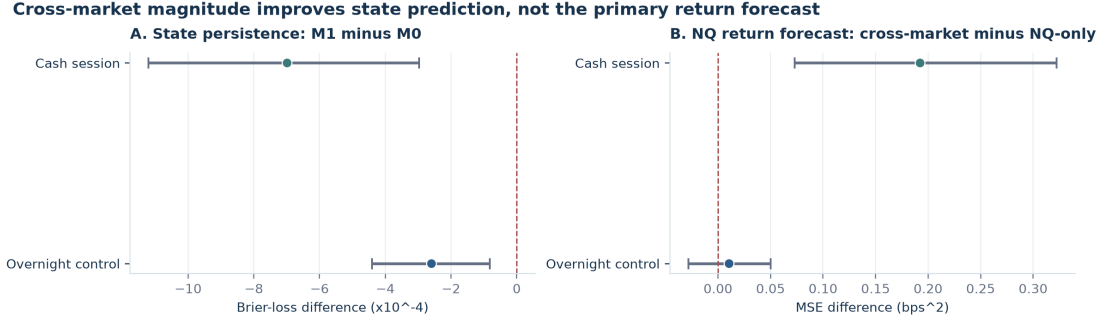


Figure 2: Primary five-minute paired loss differences. Points are out-of-fold means; horizontal bars are 95% complete-session bootstrap intervals. Negative values favor the expanded cross-market model. The panels use different loss units.

6.3 Incremental return value

The return sample contains 11,364 events from the same 264 validation sessions. Neither Ridge model beats the fold training-mean benchmark, and the expanded information set is worse than NQ-only under both registered losses.

Table 4: Five-minute signed NQ return forecast

Model	MSE (bps ²)	RMSE (bps)	MAE (bps)	OOS R^2
Fold training mean	134.831	11.612	7.080	0.0000
NQ-only Ridge	135.101	11.623	7.092	-0.0020
Cross-market Ridge	135.294	11.632	7.099	-0.0034
Cross-market – NQ-only	+0.1924	—	+0.0073	—
95% block interval	[+0.0728, +0.3222]	—	[+0.0010, +0.0135]	—

Notes: Lower forecast loss is better. The positive paired differences favor NQ-only. OOS R^2 is relative to the fold-local training-mean forecast.

The registered incremental-return criterion is therefore not met. Strong shrinkage is itself informative: $\alpha = 1000$ is selected in 9 of 12 NQ-only folds and 8 of 12 cross-market folds. The nested process generally prefers predictions close to the local mean instead of large feature-driven variation.

7 Robustness and negative control

At one-minute resolution, $M_1 - M_0$ Brier loss is -0.000297 with 95% interval $[-0.000418, -0.000173]$ across 67,483 events. The five-minute economic endpoint remains fixed, however, and the cross-market minus NQ-only return MSE difference is $+0.0031 \text{ bps}^2$ with interval $[-0.1065, +0.0962]$. The finer representation therefore supports the narrow state result but does not rescue the return hypothesis.

The overnight state sample contains 36,230 observations. Its $M_1 - M_0$ Brier difference is -0.000259 with interval $[-0.000440, -0.000081]$. Cash-session specificity is therefore not supported. The overnight return comparison remains economically unhelpful: cross-market minus NQ-only MSE is $+0.0106 \text{ bps}^2$ with interval $[-0.0283, +0.0502]$, while MAE worsens and both Ridge models trail the fold mean.

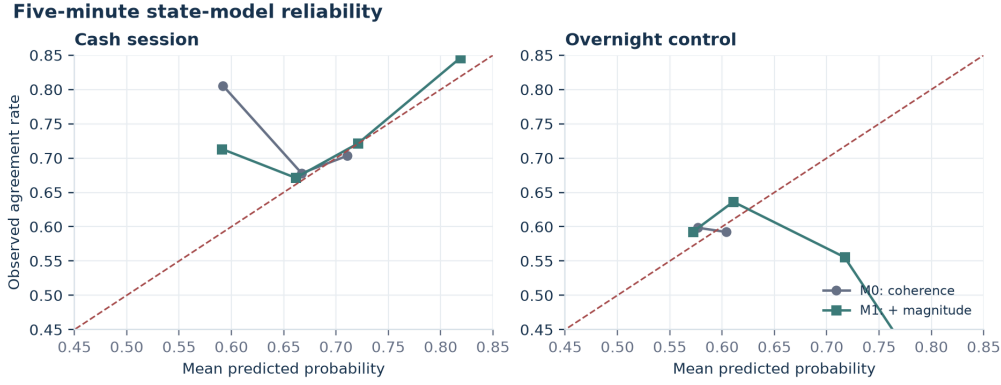


Figure 3: Reliability diagrams for the five-minute cash-session state model and the overnight negative control. Empty probability bins are omitted from the lines but retained in the committed calibration tables.

8 Limitations and interpretation

The evidence is deliberately narrower than a typical trading-strategy claim.

- The sample has no pre-ETP history, so the motivating market-integration narrative is neither identified nor tested causally.
- No pristine holdout exists. Intervals condition on the fitted out-of-fold predictions and do not quantify uncertainty from repeating the entire research process on an independent history.
- NQ is a partial stitched series. The complete historical roll-selection rule and contract-level lineage cannot be reconstructed.
- Missing sessions are systematic; notably few Thursdays survive. External validity cannot assume missingness at random.
- NQ price behavior is not MNQ execution evidence. No transaction-cost, liquidity, slippage, position-sizing, or deployment claim is made.

The state result is statistically stable within the development sample but practically small. Its repetition overnight weakens a cash-session-specific interpretation. Most importantly, the return test demonstrates that improved state classification need not translate into improved conditional-mean return prediction. Opening an entry/exit search after that result would increase researcher degrees of freedom without an economic basis.

9 Conclusion

Volatility-adjusted joint magnitude adds a small, repeatable amount of information about whether synchronized BTC and NQ candle-body directions persist into the next bar. It does not improve the registered five-minute NQ return forecast over an NQ-only baseline. The appropriate conclusion is conditional state dependence without demonstrated alpha.

This negative economic result is useful: it rejects the simplest tradability route while preserving a precisely defined state variable for future work. A confirmatory extension would require complete contract-level data, a genuinely untouched period, and a new protocol fixed before any nonlinear model, event filter, or coherence-decay policy is evaluated.

Reproducibility and credit

Code, configurations, trial ledgers, aggregate results, figures, and SHA-256 manifests are versioned in the public repository. Raw proprietary NQ rows remain local. The research question originated in earlier collaborative coursework with Alex Samuseva; the present repository is a clean redesign and does not treat legacy notebook results as evidence. No license has been selected pending a joint authorship decision.

References

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